

torch.cat#

<https://pytorch.org/docs/stable/generated/torch.cat.html>

Description:

Concatenates the given sequence of tensors in tensors in the given dimension. All tensors must either have the same shape (except in the concatenating dimension) or be a 1-D empty tensor with size (0,). `torch.cat()` can be seen as an inverse operation for `torch.split()` and `torch.chunk()`. `torch.cat()` can be best understood via examples. `torch.stack()` concatenates the given sequence along a new dimension.

Function Signature

```
torch.cat(tensors, dim=0, *, out=None) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> x = torch.randn(2, 3)
>>> x
tensor([[ 0.6580, -1.0969, -0.4614],
        [-0.1034, -0.5790,  0.1497]])
>>> torch.cat((x, x, x), 0)
tensor([[ 0.6580, -1.0969, -0.4614],
        [-0.1034, -0.5790,  0.1497],
        [ 0.6580, -1.0969, -0.4614],
        [-0.1034, -0.5790,  0.1497],
        [ 0.6580, -1.0969, -0.4614],
        [-0.1034, -0.5790,  0.1497]])
>>> torch.cat((x, x, x), 1)
tensor([[ 0.6580, -1.0969, -0.4614,  0.6580, -1.0969, -0.4614,
          0.6580,
          -1.0969, -0.4614],
        [-0.1034, -0.5790,  0.1497, -0.1034, -0.5790,  0.1497,
          -0.1034,
          -0.5790,  0.1497]])
```

Notes & Warnings

NOTE

See also `torch.stack()` concatenates the given sequence along a new dimension.

Detailed Documentation

Documentation

Concatenates the given sequence of tensors in tensors in the given dimension. All tensors must either have the same shape (except in the concatenating dimension) or be a 1-D empty tensor with size (0,).

`torch.cat()` can be seen as an inverse operation for `torch.split()` and `torch.chunk()`.

`torch.cat()` can be best understood via examples.

`torch.stack()` concatenates the given sequence along a new dimension.

tensors (sequence of Tensors) – Non-empty tensors provided must have the same shape, except in the cat dimension.

dim (int, optional) – the dimension over which the tensors are concatenated

out (Tensor, optional) – the output tensor.
